SHEAF AND LIMIT PRESERVING PROPERTIES OF THE FUNCTORS OF POINTS WHERE A MODULE MAP IS INJECTIVE, SURJECTIVE, BIJECTIVE OR ZERO

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

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Allowed standard prerequisites (editorial).

- The situation and functors recording injectivity, surjectivity, bijectivity and vanishing of a module map. Same chapter flat.tex, situation labelled situation-iso (statement, proof line 5920). Fixes the four functors F_iso, F_inj, F_surj and F_zero whose sheaf and limit preserving properties are asserted.
- Flatness of an fpqc covering and faithful flatness of the induced local maps. Topologies, label topologies-lemma-fpqc (proof line 5937); Algebra, label algebra-lemma-local-flat-ff (proof line 5942). Shows that the pullback covering of X_T is an fpqc covering, so every point has a point of some member over it, and that the resulting local map of stalks is faithfully flat, which is what makes injectivity, surjectivity, bijectivity and vanishing equivalent upstairs and downstairs.
- The algebraic colimit statements for module maps and reduction of finite type and finite presentation modules. Algebra, label algebra-lemma-module-map-property-in-colimit, parts (1), (2) and (3), invoked at proof lines 5975, 5987 and 6004; Properties, labels properties-lemma-finite-type-module (proof line 5966) and properties-lemma-finite-presentation-module (proof line 5994). Each of the limit preserving parts is reduced to the corresponding algebraic statement that surjectivity, vanishing or bijectivity of a base changed map descends to some stage of a directed colimit, and the reduction from sheaves to modules uses the affine local description of a finite type or finite presentation module.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 00HR: sources/stacks-chow/tags.txt, source lines 653–653. Navigation pointer for \ref{algebra-lemma-local-flat-ff} used inside the counted span (line 5942); the argument that consumes it is retained in the span.

Source reference. Original lemma 05LI: sources/stacks-chow/tags.txt, source lines 7237–7237. Navigation pointer for \ref{algebra-lemma-module-map-property-in-colimit} used inside the counted span (line 5975); the argument that consumes it is retained in the span.

Source reference. Original lemma 01PC: sources/stacks-chow/tags.txt, source lines 2225–2225. Navigation pointer for \ref{properties-lemma-finite-presentation-module} used inside the counted span (line 5994); the argument that consumes it is retained in the span.

Source reference. Original lemma 01PB: sources/stacks-chow/tags.txt, source lines 2224–2224. Navigation pointer for \ref{properties-lemma-finite-type-module} used inside the counted span (line 5966); the argument that consumes it is retained in the span.

Source reference. Original lemma 05MH: sources/stacks-flat/pinned-original-source.tex, source lines 5896–5896. Navigation pointer for \ref{situation-iso} used inside the counted span (line 5920); the argument that consumes it is retained in the span.

Source reference. Original lemma 022D: sources/stacks-chow/tags.txt, source lines 2702–2702. Navigation pointer for \ref{topologies-lemma-fpqc} used inside the counted span (line 5937); the argument that consumes it is retained in the span.

Lemma 0.1. *In Situation 05MH.*

(1) *Each of the functors F_{iso}, F_{inj}, F_{surj}, F_{zero} satisfies the sheaf property for the fpqc topology.*
(2) *If f is quasi-compact and $\mathcal{G}$ is of finite type, then F_{surj} is limit preserving.*
(3) *If f is quasi-compact and $\mathcal{F}$ of finite type, then F_{zero} is limit preserving.*
(4) *If f is quasi-compact, $\mathcal{F}$ is of finite type, and $\mathcal{G}$ is of finite presentation, then F_{iso} is limit preserving.*

Proof. Let $\{T_i \to T\}_{i\in I}$ be an fpqc covering of schemes over S. Set $X_i = X_{T_i} = X \times_S T_i$ and $u_i = u_{T_i}$. Note that $\{X_i \to X_T\}_{i\in I}$ is an fpqc covering of X_T, see Topologies, Lemma 022D. In particular, for every $x \in X_T$ there exists an $i \in I$ and an $x_i \in X_i$ mapping to x. Since $\mathcal{O}_{X_T,x} \to \mathcal{O}_{X_i,x_i}$ is flat, hence faithfully flat (see Algebra, Lemma 00HR) we conclude that $(u_i)_{x_i}$ is injective, surjective, bijective, or zero if and only if $(u_T)_x$ is injective, surjective, bijective, or zero. Whence part (1) of the lemma.

Proof of (2). Assume f quasi-compact and $\mathcal{G}$ of finite type. Let $T = \lim_{i\in I} T_i$ be a directed limit of affine S-schemes and assume that u_T is surjective. Set $X_i = X_{T_i} = X \times_S T_i$ and $u_i = u_{T_i} : \mathcal{F}_i = \mathcal{F}_{T_i} \to \mathcal{G}_i = \mathcal{G}_{T_i}$. To prove part (2) we have to show that u_i is surjective for some i. Pick $i_0 \in I$ and replace I by $\{i \mid i \geq i_0\}$. Since f is quasi-compact the scheme X_{i_0} is quasi-compact. Hence we may choose affine opens $W_1, \ldots, W_m \subset X$ and an affine open covering $X_{i_0} = U_{1,i_0} \cup \ldots \cup U_{m,i_0}$ such that U_{j,i_0} maps into W_j under the projection morphism $X_{i_0} \to X$. For any $i \in I$ let $U_{j,i}$ be the inverse image of U_{j,i_0}. Setting $U_j = \lim_i U_{j,i}$ we see that $X_T = U_1 \cup \ldots \cup U_m$ is an affine open covering of X_T. Now it suffices to show, for a given $j \in \{1, \ldots, m\}$ that $u_i|_{U_{j,i}}$ is surjective for some $i = i(j) \in I$. Using Properties, Lemma 01PB this translates into the following algebra problem: Let A be a ring and let $u : M \to N$ be an A-module map. Suppose that $R = \operatorname{colim}_{i\in I} R_i$ is a directed colimit of A-algebras. If N is a finite A-module and if $u \otimes 1 : M \otimes_A R \to N \otimes_A R$ is surjective,

then for some i the map $u \otimes 1 : M \otimes_A R_i \to N \otimes_A R_i$ is surjective. This is Algebra, Lemma 05LI part (2).

Proof of (3). Exactly the same arguments as given in the proof of (2) reduces this to the following algebra problem: Let A be a ring and let $u : M \to N$ be an A-module map. Suppose that $R = \operatorname{colim}_{i \in I} R_i$ is a directed colimit of A-algebras. If M is a finite A-module and if $u \otimes 1 : M \otimes_A R \to N \otimes_A R$ is zero, then for some i the map $u \otimes 1 : M \otimes_A R_i \to N \otimes_A R_i$ is zero. This is Algebra, Lemma 05LI part (1).

Proof of (4). Assume f quasi-compact and $\mathcal{F}, \mathcal{G}$ of finite presentation. Arguing in exactly the same manner as in the previous paragraph (using in addition also Properties, Lemma 01PC) part (3) translates into the following algebra statement: Let A be a ring and let $u : M \to N$ be an A-module map. Suppose that $R = \operatorname{colim}_{i \in I} R_i$ is a directed colimit of A-algebras. Assume M is a finite A-module, N is a finitely presented A-module, and $u \otimes 1 : M \otimes_A R \to N \otimes_A R$ is an isomorphism. Then for some i the map $u \otimes 1 : M \otimes_A R_i \to N \otimes_A R_i$ is an isomorphism. This is Algebra, Lemma 05LI part (3). □